

Contents

Introduction	3
Why these disorders?.....	4
What is neurodiversity	4
What disorders are we covering, and why?	4
Dyslexia.....	4
Dyspraxia	5
Attention Deficit Hyperactivity Disorder (ADHD)	5
Autism (autism spectrum disorder)	5
Processing disorders	6
Why is a recourse like this relevant to Cardiff University, and education as a whole? ..	7
Why these tools?	8
How did Gemini Notebook perform in our analysis?	11
Learner profiles	12
Disclaimers	14
Gemini notebook	15
What disorders can this tool help with.....	15
Autism	15
Dyslexia.....	16
ADHD.....	16
Processing disorders	17
Modern languages assessments/ tasks this tool can help with.....	18
Podcasts	18
Portfolios.....	18
Vocabulary tests	18
Desktop	19
Setting up/ signing in	19
Summarising long pdfs/ documents	20
Create quizzes and audio files from sources uploaded.....	21
Uploading lectures, using them as sources and transcribing them.....	23
Further recourses (Provided by Cardiff University)	26

References	27
Cardiff University Support services	28

Introduction

This guide explores how selected Artificial Intelligence (AI) tools can support learners with neurodevelopmental disorders, with a focus on the study of Modern Languages. As digital technologies become increasingly integrated into education, AI offers new opportunities to reduce barriers to learning, promote independence, and provide personalised support tailored to individual needs. For students with conditions such as dyslexia, autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), and other neurodevelopmental differences, AI tools can help make language learning more accessible and engaging.

Modern Language assessments often require students to demonstrate a broad range of skills, including reading, writing, listening, speaking, grammar, vocabulary acquisition, and cultural understanding. These tasks can present additional challenges for neurodivergent learners, particularly where difficulties with processing, organisation, memory, concentration, or communication are involved. AI technologies can provide practical assistance by offering real-time feedback, personalised learning pathways, speech-to-text and text-to-speech capabilities, translation support, vocabulary reinforcement, and study organisation tools.

The intended audience for this guide is undergraduate students at Cardiff University, with a focus on those studying Modern Languages. This guide covers a wide range of functionality provided by selected AI tools, how to use them, and how they can help Modern Languages students with task preparation. This guide is a baseline for how AI tools can be used, and in no way covers every single use of every single tool covered, just ways in which we have identified could help students. This guide provides a baseline overview of how AI tools may be used to support neurodivergent students. It does not seek to cover every possible application of the tools included but rather highlights selected uses that were identified as potentially beneficial across different aspects of the student experience. The development of the guide was further informed by feedback from volunteer students within the Modern Languages (MLANG) community, whose reflections and lived experiences helped refine its content, relevance, and accessibility.

AI tools should be viewed as supplementary support mechanisms rather than replacements for existing reasonable adjustments, specialist support services, or inclusive teaching practices. The effectiveness of any tool will vary according to individual needs, preferences, and contexts, and students should be encouraged to identify the approaches that best support their learning.

Why these disorders?

What is neurodiversity

Neurodiversity is the concept of that there is natural variation in the ways that the human brain works, signifying that there is no one “correct” way to think, learn, or process information.

“Neurodiversity is the concept that there is natural variation in how people’s brains work, with no single “correct” way, and that a wide range of perceptions and responses to the world should be accepted and encouraged” – (Child Mind Institute, 2026).

What disorders are we covering, and why?

Dyslexia

Evidence

Dyslexia effects reading accuracy, spelling, phonological processing and vocabulary. These difficulties can impact reading comprehension, grammar, and written language production in Modern Languages. (University of Strathclyde, n.d.)

Rationale for inclusion

Modern Language relies heavily on reading, writing, spelling, and pronunciation. AI tools that can perform text-to-speech, speech recognition, and grammar support that can provide huge benefits for those who struggle.

Possible link to assessments/ tasks

- Writing exams
- Reading exams
- Vocabulary
- Pronunciation activities

Dyspraxia

Evidence

Dyspraxia can affect motor coordination, planning and organisation, handwriting speed and notetaking. People with dyspraxia may also struggle with planning (e.g. creating a plan for tasks and projects)

Rationale for inclusion

Written tasks and language learning activities often require extensive notetaking and written production, AI based speech-to-text may reduce these barriers. AI tools can also be used to effectively plan tasks and create timelines.

Possible link to assessment/tasks

- Notetaking
- Written Tasks
- Project Planning

Attention Deficit Hyperactivity Disorder (ADHD)

Evidence

ADHD can affect concentration, working memory, time management and task completion. Learners may struggle to remember vocabulary, organise revision, or sustain attention span during lengthy lessons and tasks. (Royal Borough of Windsor and Maidenhead, n.d.)

Rationale for inclusion

Modern Languages tasks require sustained focus and structured responses. AI tools can support planning, revision planning, resource organisation, and personalised learning.

Possible links to assessments/ tasks

- Writing task
- Listening tasks
- Revision and coursework management

Autism (autism spectrum disorder)

Evidence

Autistic learners may experience challenges with social communication, interpreting figurative language, or understanding contextual clues. Language learning often involves cultural interpretation and communication skills. (Reachout ASC (n.d.)).

Rationale for inclusion

Modern Languages requires both linguistic and social communication competencies. AI tools can provide structured explanations, conversational practise and personalised learning environments.

Possible links to assessments/ tasks

- Speaking assessments
- Cultural interpretation tasks
- Group tasks

Processing disorders

Evidence

Processing disorders can affect auditory, visual, and speed of processing. This means that tasks such as reading, listening, following instructions, vocabulary retention, and language comprehension may be challenging, particularly in the study of Modern Languages. (National Center for Learning Disabilities (n.d.))

Rationale for inclusion

Modern Language Learning requires learners to quickly process spoken language and written information, remember new vocabulary, and interpret meaning from context. AI tools that provide text-to-speech, content summarisation, and personalised pacing can help reduce barriers for learners with processing difficulties

Possible link to assessments/ tasks

- Writing exams
- Reading exams
- Vocabulary
- Multi step instructions
- Note taking

Why is a recourse like this relevant to Cardiff University, and education as a whole?

Inclusive education and recognition of learners at Cardiff University with neurodevelopmental disorders is very important. Neurodivergence is real and impacts people in all different areas of life. Cardiff University has 7% (1,980) of students that have disclosed that they are neurodivergent as of the 25/26 academic year, along with 7.8% (40) of MLANG students that have also disclosed the fact they are neurodivergent.

With this number of students that have disclosed that they are neurodivergent, it is still very common for people to get to the point of being in the first year of their university course, unaware that they are neurodivergent, and therefore do not get the support they need. Promoting this topic is important, as it will allow for people to get support they did not know they needed, helping further contribute to equality in education.

Statistics from higher education state that neurodiversity is increasingly important in the sector, with recent statistics showing that 22% of UK students have a neurodivergent diagnosis, with up to 28% identifying as neurodivergent in some way. This contributes to an overrepresentation of neurodivergence among students compared to the general population, of which only 15-20% are believed to be neurodivergent (Eves, 2026). This highlights the growing importance of neurodiversity in higher education and reinforces the need for targeted strategies and resources that enable neurodivergent students to thrive academically

Why these tools?

We conducted a comprehensive review of a wide range of tools to identify those most suitable for neurodivergent students at Cardiff University. The evaluation began with an initial shortlist of 11 tools, which were assessed against a set of rigorous criteria to determine their suitability and effectiveness. The criteria used in the evaluation are outlined below:

Evaluation criteria and rating of selected AI tools

What we are evaluating these tools for:

Criterion	Definition	Indicators	Rating (1-5)
Accessibility	The extent to which this tool is accessible for the targeted students with neurodevelopmental disorders	Text-to-speech, clear language, keyboard accessibility, simple interface, adjustable text size, captions.	1 = Poor 5 = Outstanding
Personalisation	The features provided which allow the users to change the app, to promote a more user specific experience	Tailored explanations, adjustable difficulty, personalised feedback, response customisation.	1 = Poor 5 = Outstanding
Cognitive load and management	The extent that the tool can help reduce unnecessary mental effort	Step-by-step guidance, concise responses, chunking of information, minimal distractions.	1 = Poor 5 = Outstanding
Pedagogical Usefulness	How well the AI tool supports meaningful learning and problem solving.	Accurate explanations, scaffolding, constructive feedback, encourages critical thinking.	1 = Poor 5 = Outstanding

The criteria for scoring:

Score	Meaning
1	Does not meet the criteria, not fit or purpose
2	Meets the criteria to a limited extent, improvements would be needed
3	Meets the criteria moderately, basic use possible
4	Meets the criteria, intended use very possible
5	Meets the criteria fully, intended use is excellently provided.

First, we started creating a series of relevant categories, aligned to features in an app that would be useful to our target audience. We then created a criterion for scoring across characteristics of every category we have defined.

Below are the scoring criteria:

Accessibility:

Indicator	What to look for	Rating 0 – 5
Clear and readable interface	Clean layout, readable fonts, minimal clutter	
Accessible communication	Uses plain language and can simplify explanations	
Multiple ways to interact	Supports text, voice, images, or other input/output methods	
Computability with accessibility features	Works with screen readers, keyboard navigation, text-to-speech, etc.	
Sensory accessibility	Avoids overwhelming visuals or unnecessary distractions	
Total		Points average

Personalisation:

Indicator	What to look for	Points 0 – 5
Adaptive explanations	can simplify, expand, or rephrase content	
Adjustable response style	User can request different formats, lengths, or levels of detail	
Personalised feedback	Feedback reflects the learner's responses rather than generic advice	
Supports individual learning pace	Learners can revisit, repeat, or build on previous explanations	
Flexibility	Can generate examples, quizzes, summaries, or alternative explanations	
Total		Points average

Cognitive load management

Indicator	What to look for	Points 0 – 5
Chunking information	Breaks information into manageable sections	
Step by step guidance	Explains tasks sequentially	
Concise explanations	Avoids unnecessary complexity or information overload	
Organisation	Uses headings, bullet points, summaries, or logical structure	
Attention support	Keeps responses focused and avoids irrelevant information	
Total		Points

Pedagogical usefulness

Indicator	What to look for	Points 0 – 5
Concept explanation	Explains reasoning rather than only giving answers	
Constructive feedback	Identifies mistakes and explains how to improve	
Encourages critical thinking	Prompts learners to think rather than simply copy	
Supports independent learning	Helps learners solve problems themselves	
Learning reinforcement	Provides examples, practice questions, summaries, or revision support	
Total		Points

From this point, we made note of an average score from a section, and add it to every other average score to get a total score for the app. From this we were able to see how every tool we analysed compared with each other, and which tools aligned best with the needs of students with neurodevelopmental disorders.

Below is the selection of tools we analysed, along with the task they were best aligned for:

Reading	Writing	Listening	Speaking
• Perplexity • Notebook LM • Knowt • Language tool • Wolfram alpha • Photomath	• Microsoft Copilot (w/ voice) • Quillbot	• Google Gemini (w/ live mode)	• Google Gemini (w/ live mode)

Below is the result from our analysis:

Final scores:

AI tool	Score
Perplexity	16.4 / 20
Notebook LM	15.6 / 20
Knowt	15.6 / 20
Quillbot	16.5 / 20
Language tool	12.9 / 20
Wolfram alpha	10.8 / 20
Photo math	11.8 / 20
Microsoft Copilot (w/ voice)	16.2 / 20
Google Gemini (w/ live mode)	16.5 / 20

Score breakdown:

Score range	Description
0 – 8.9	Not fit for purpose
9 – 11.9	Fit for purpose but may have limited use case.
12 – 13.9	Fit for purpose
14 – 15.9	Fit for purpose, has great features, and a good use case
16 or more	Fit for purpose, has excellent features, and a great use case.

From this we were able to create a further shortlist of tools:

Cardiff University Provide Gemini Notebook, Google Gemini, and Microsoft Copilot. This also helped us when deciding which tools to shortlist.

The final shortlisted tools:

Reading	Writing	Listening and speaking
• Perplexity • Notebook LM • Language tool	• Microsoft Copilot (w/ voice) • Quillbot	• Google Gemini (w/ live mode)

How did Gemini Notebook perform in our analysis?

Gemini Notebook

Accessibility:

Indicator	What to look for	Rating 0 – 5
Clear and readable interface	Clean layout, readable fonts, minimal clutter	4
Accessible communication	Uses plain language and can simplify explanations	4
Multiple ways to interact	supports text, voice, images, or other input/output methods	2
Computability with accessibility features	Works with screen readers, keyboard navigation, text-to-speech, etc.	2
Sensory accessibility	Avoids overwhelming visuals or unnecessary distractions	4
Total		Points average: 3.2

Personalisation:

Indicator	What to look for	Points 0 – 5
Adaptive explanations	can simplify, expand, or rephrase content	3
Adjustable response style	User can request different formats, lengths, or levels of detail	4
Personalised feedback	Feedback reflects the learner's responses rather than generic advice	4
Supports individual learning pace	Learners can revisit, repeat, or build on previous explanations	4
Flexibility	Can generate examples, quizzes, summaries, or alternative explanations	5
Total		Points average: 4

Cognitive load management:

Indicator	What to look for	Points 0 – 5
Chunking information	Breaks information into manageable sections	4
Step by step guidance	Explains tasks sequentially	4
Concise explanations	Avoids unnecessary complexity or information overload	4
Organisation	Uses headings, bullet points, summaries, or logical structure	4
Attention support	Keeps responses focused and avoids irrelevant information	5
Total		Points average: 4.2

Pedagogical usefulness:

Indicator	What to look for	Points 0 – 5
Concept explanation	Explains reasoning rather than only giving answers	4
Constructive feedback	Identifies mistakes and explains how to improve	4
Encourages critical thinking	Prompts learners to think rather than simply copy	4
Supports independent learning	Helps learners solve problems themselves	4
Learning reinforcement	Provides examples, practice questions, summaries, or revision support	5
Total		Points average: 4.2

SCORE: 15.6 / 20

Learner profiles

Every student experience learning differently. To illustrate how neurodiversity can shape academic experiences, we've created a series of learner profiles based on common student experiences that learners that are neurodivergent may have. You can find these learner profiles below:

Profile 1: Hyperactive ADHD (Adam)

- **Demographics:** 18, 1st Year BA Modern Languages (English native, learning Spanish).
- **Core Profile:** Outgoing and highly creative, but struggles with executive dysfunction, time management, hyperfixation shifts, and finishing multi-step tasks.

Profile Dimension	Details & Strategies
Strengths	High energy, creativity, hyperfocus capacity, enthusiasm, strong social/conversational potential.
Academic Friction	Initiating unstructured study; managing long-term essay deadlines; remaining focused during long grammar lectures.
AI Matching Strategy	Use AI to externalise executive function and gamify micro-tasks.
Concrete AI Workflows	• Microsoft Copilot: "Break down my 2,000-word Spanish literature essay into 5 daily micro-tasks. Put deadlines on each and format as a checklist for Microsoft To-Do." • LanguageTool: Passively catches grammar/spelling errors while drafting without requiring dedicated editing sessions. • Perplexity AI: Focuses research queries to avoid falling down irrelevant internet "rabbit holes."
Assessment-Targeted Tasks	• Targeted Task (Coursework Essays): Prompt Copilot to build a structured 7-day writing roadmap with micro-deadlines. • Targeted Task (Grammar & Syntax Exams): Use LanguageTool for immediate feedback loops while practicing past papers. • Targeted Task (Literature & Culture Projects): Use Perplexity to gather verified, in-scope academic sources quickly without distraction.
Pedagogical Guardrail	Set strict time limits on AI interactions so hyperfocus isn't spent tweaking AI prompts instead of studying.

Profile 2: Inattentive ADHD (Amelia)

- **Demographics:** 18, 1st Year BA Modern Languages (English native, learning French).
- **Core Profile:** Deep thinker with strong pattern recognition but paralysed by high activation energy (task initiation), low executive resilience, and anxiety around foreign language speaking.

Profile Dimension	Details & Strategies
Strengths	Deep thinking, empathy, curiosity, conceptual pattern recognition.
Academic Friction	High task paralysis when staring at blank pages; anxiety surrounding real-time oral speaking in front of peers.
AI Matching Strategy	Use AI to lower initiation friction and provide low-stakes, judgement-free practice.
Concrete AI Workflows	• Google Gemini (Live Mode) / Copilot Voice: Low-stakes practice holding basic French conversations without peer-judgment or performance anxiety. • NotebookLM: Upload module reading PDFs to generate 3-bullet point summaries, reducing cognitive load before diving into full texts.
Assessment-Targeted Tasks	• Targeted Task (Oral Speaking Exams): Practice holding low-stakes, conversational roleplays with Gemini Live Mode to lower anxiety prior to real oral exams. • Targeted Task (Reading Comprehension Exams): Use NotebookLM to generate initial core summaries of dense set texts before reading the full material. • Targeted Task (Essay Planning): Prompt Copilot to generate 3 different starter outlines to overcome blank-page initiation paralysis.
Pedagogical Guardrail	Ensure she uses conversational AI as a stepping stone to build confidence for real human interaction, not as a permanent substitute.

Student Guide | Gemini Notebook

Profile 3: Dyslexia (Samuel)

- **Demographics:** 19, 1st Year BA Modern Languages (Welsh national, limited Welsh, passionate about new languages).
- **Core Profile:** Resilient and creative, but experiences severe working memory load and processing friction when decoding written text, symbols, and spelling.

Profile Dimension	Details & Strategies
Strengths	Resilience, verbal communication, creative problem solving.
Academic Friction	Reading dense foreign text; accurate spelling/grammar in target languages; decoding written exam materials.
AI Matching Strategy	Transition from text-heavy interaction to multimodal (audio/visual) learning .
Concrete AI Workflows	• QuillBot & LanguageTool: Dual-proofreading layer to catch subtle grammatical inflections and orthographic errors. • Google Gemini (Multimodal Vision): Snap a photo of a printed Spanish text or worksheet → request audio read-back and simplified translation breakdowns. • NotebookLM: Convert dense syllabus PDFs into automatically generated audio overviews/podcasts and practice quizzes.
Assessment-Targeted Tasks	• Targeted Task (Written Composition & Reports): Use QuillBot and LanguageTool as a final proofreading step to catch spelling and grammar slips. • Targeted Task (Translation & Text Analysis): Snap photos of physical texts using Gemini to hear them read aloud while reviewing syntax. • Targeted Task (Revision & Knowledge Checks): Convert lecture slides and notes into automated practice quizzes using NotebookLM.
Pedagogical Guardrail	Require double-checking QuillBot output against class grammar rules, as AI suggestions can occasionally alter target-language nuances.

Profile 5: Slow Processing Speed (Declan)

- **Demographics:** 18, 1st Year BA Modern Languages (Bilingual Welsh/English).
- **Core Profile:** Accurate and deep thinker when given time but struggles to keep up with fast-paced live lectures and rapid spoken audio.

Profile Dimension	Details & Strategies
Strengths	High accuracy, deep conceptual understanding, methodical work habits.
Academic Friction	Taking real-time notes during lectures; rapid oral comprehension exercises; fast-paced seminar discussions.
AI Matching Strategy	Use AI for asynchronous pacing control and information distillation .
Concrete AI Workflows	• Copilot / Gemini Voice: Upload audio/video recordings of lectures to generate full transcripts, allowing Declan to review at his own pace. • NotebookLM: "Summarise this lecture transcript into core key takeaways and a 10-word vocabulary list to reduce reading volume."
Assessment-Targeted Tasks	• Targeted Task (Listening Comprehension Tests): Generate transcripts of practice audio using Copilot/Gemini to study sentence structure asynchronously. • Targeted Task (End-of-Term Exam Preparation): Use NotebookLM to condense voluminous term readings into high-priority core summary sheets. • Targeted Task (Seminar Preparation): Use NotebookLM to create pre-class key vocabulary lists to reduce real-time cognitive processing load.
Pedagogical Guardrail	Ensure audio transcripts are reviewed alongside lecture slides to verify technical term accuracy.

Profile 4: Autistic Learner (Austin)

- **Demographics:** 18, 1st Year BSc Computer Science (Welsh national, English speaking).
- **Core Profile:** Highly analytical with deep attention to detail, but vulnerable to sensory overload, unpredictable environment shifts, and figurative/ambiguous language.

Profile Dimension	Details & Strategies
Strengths	Logical thinking, detail accuracy, systematic problem solving, high reliability.
Academic Friction	Interpreting non-literal assignment prompts; managing sensory overload during crowded lectures; navigating group work.
AI Matching Strategy	Use AI as a clarifier to strip away ambiguity and create explicit structure .
Concrete AI Workflows	• Copilot / NotebookLM: Direct prompt: "Rewrite these assignment instructions into literal, step-by-step instructions. Remove metaphorical language and list explicit deliverables." • NotebookLM: Converts lecture notes into structured visual diagrams, explicit glossaries, and factual Q&As to study in a quiet, low-sensory setting.
Assessment-Targeted Tasks	• Targeted Task (Interpreting Assignment Briefs): Paste module rubrics into Copilot to generate explicit, step-by-step deliverable checklists. • Targeted Task (Theoretical/Conceptual Exams): Use NotebookLM to compile literal, factual glossaries and visual diagrams from course files. • Targeted Task (Group Project Planning): Use Copilot to clearly divide tasks and write unambiguous communication templates for team members.
Pedagogical Guardrail	Explicitly define parameters for AI use so Austin understands exact academic integrity boundary lines.

Profile 6: Auditory Processing Disorder - APD (Sebastian)

- **Demographics:** 18, 1st Year BA Modern Languages (English native, limited Spanish).
- **Core Profile:** Visual learner who struggles to distinguish and process spoken auditory input, leading to fatigue in listening exercises.

Profile Dimension	Details & Strategies
Strengths	Visual processing capacity, spatial learning, resilience, hands-on learning.
Academic Friction	Foreign language listening exams; auditory lectures; background noise in oral seminars.
AI Matching Strategy	Convert auditory input into visual text and visual diagrams .
Concrete AI Workflows	• Google Gemini / Copilot Live: Live transcription of spoken class audio into formatted, real-time written text. • NotebookLM: Input audio recordings to produce written study summaries, mind maps, and flashcard sets, completely bypassing auditory fatigue.
Assessment-Targeted Tasks	• Targeted Task (Audio/Listening Assessments): Practice with Gemini by pairing spoken audio outputs directly with real-time screen transcription. • Targeted Task (Lecture Content Mastery): Process lecture audio through NotebookLM to create visual mind maps and text notes. • Targeted Task (Vocabulary Building): Convert audio flashcards into visual/written vocabulary matrices.
Pedagogical Guardrail	Target language phonology still needs practice; use synchronised text-and-audio playback tools rather than cutting out listening entirely.

Disclaimers

- Always use your Cardiff University email address to access any AI tools recommended by the University. Where available, use the University-provided or University-approved versions of these tools. This helps ensure that your data is protected under Cardiff University's security, privacy and data protection arrangements.
- Avoid sharing personal, confidential, or sensitive information with AI tools whenever possible. This includes information about yourself, other students, staff members or any data that could identify an individual.
- The advice provided in this guide, and in any related University guidance, is intended to support your learning and should be considered guidance rather than a set of mandatory rules. You should always use your own judgement and follow any instructions provided by your module leaders or the University.
- AI-generated content is not always accurate. AI tools can sometimes produce incorrect, misleading, incomplete, or fabricated information (often referred to as "hallucinations"). You should carefully review, verify and critically evaluate all AI-generated outputs before using them in your studies.
- Use AI as a learning aid, not as a substitute for your own work. AI can help you understand concepts, organise ideas, practise language skills and receive feedback, but it should not be used to plagiarise or misrepresent work as your own. Always follow Cardiff University's policies on academic integrity and appropriate use of AI.

Gemini notebook

What disorders can this tool help with

Autism

This guide draws on research comparing the tasks that can be completed using Gemini Notebook, and its characteristics to issues students with autism may experience. Below will describe how the selected tasks have been set out further along with this guide, involving Gemini Notebook, can help autistic learners.

This guide is informed by an analysis of Gemini Notebook features and capabilities in relation to common experiences and support needs among autistic students. The following sections describe how the Gemini Notebook tasks included throughout the guide may help autistic learners engage more effectively with their studies.

Summarising long PDF's/ Documents

Autistic learners may have a harder with sensory input and may be more sensitive to sensory input when comparing to others. Research also suggests that autistic learners may struggle with ambiguous language and may benefit with description that could be classed as more "Straight to the point". Gemini Notebook helps with this, as it can summarise your learning recourses, into easier to digest language, and in a smaller formfactor. This can help if a learner wishes to use more than one source at the same time.

Uploading lectures, using them as a source and transcribing them

Environments such as lectures, or any other public place, may be harder for a learner with autism to fully concentrate in, due to the possible sensitivity they have to sensory input. Gemini notebook helps with this, as you can upload and transcript lectures easily. You can also use them as a source for any questions you need answering, by prompting Gemini Notebook. This allows an autistic learner to learn at their own pace, while being able to manage sensory input themselves.

Create quizzes and audio files and other visual or auditory recourses from sources uploaded.

Learners with autism may prefer to crate different, more visual outputs. With Gemini Notebook, using your own sources, you can create visual resources, such as graphics and quizzes. This may help limit sensory input, for those who may need it.

Dyslexia

This guide is informed by a comparison of Gemini Notebooks capabilities and the challenges that learners with dyslexia may experience in academic settings. The sections below explain how the Gemini Notebook tasks included throughout this guide can support learners with dyslexia in managing their studies more effectively.

From our research, we have compared tasks that can be completed Gemini Notebook, and its characteristics to issues students with dyslexia may experience. Below will describe how the following tasks we have set out further along with this guide, involving Gemini Notebook, can help learners with dyslexia.

Summarising long PDF's/ Documents

Learners with dyslexia may benefit from minimised reading. Dyslexic learners typically have greater difficulty reading. Gemini notebook can help with this, by summarising multiple sources into section of information, meaning that the process of reading text-based information can be more efficient for those who struggle.

ADHD

This guide is informed by a comparison of Gemini Notebook capabilities and the challenges that learners with ADHD may experience in academic settings. The sections below explain how the Gemini Notebook tasks included throughout this guide can support learners with ADHD in managing their studies more effectively.

From our research, we have compared tasks that can be completed with Gemini Notebook, and its characteristics to issues students with ADHD may experience. Below will describe how the following tasks we have set out further along with this guide, involving Gemini Notebook, can help learners with ADHD.

Summarising long PDF's/ Documents

Being able to summarise long sources of information may be beneficial to those with ADHD. Learners with ADHD may benefit from summarised information as this may help reduce the mental resilience, researching and/or revising, and prolong concentration.

Uploading lectures, using them as a source and transcribing them

It may be hard for an individual with ADHD to stay concentrated during a lecture at times or at least may benefit from being able to efficiently recap. This can be done easily with Gemini Notebook, as uploading and summarising different learning sources, that can be a full lecture recording, is simple. This information can also be presented in accordance to the learners' personal preferences, with the wide range of learning recourses that Gemini Notebook offers.

Create quizzes and audio files and other visual or auditory recourses from sources uploaded.

Research reveals that learners with ADHD may benefit from limiting prolonged tedious tasks. This can be done with Gemini Notebook. Gemini notebook allows you to create personalised learning recourses, from uploaded sources. There are a wide range of revision/ learning tasks that Gemini notebook can provide, using information from your sources, such as quizzes and audio-based recourses. This can help learners with ADHD, as this can make learning more engaging, and require less mental resilience, when comparing to reading a long report.

Processing disorders

This guide is informed by a comparison of Gemini Notebook capabilities and the challenges that learners with processing disorders may experience in academic settings. The sections below explain how the Gemini Notebook tasks included throughout this guide can support learners with processing disorders in managing their studies more effectively.

From our research, we have compared tasks that can be completed with Gemini Notebook, and its characteristics to issues students with processing disorders may experience. Below will describe how the following tasks we have set out further along with this guide, involving Gemini Notebook can help learners with processing disorders.

Create quizzes and audio files and other visual or auditory recourses from sources uploaded.

Learners with processing disorders may struggle with auditory or text-based information. Gemini Notebook allows learners to visual and audio recourses, meaning that if there is a type of information formatting that learners struggle with, you can easily keep the same information, but change the way it is delivered with Gemini Notebook.

Uploading lectures, using them as sources and transcribing them

Specifically, for leaners that struggle with auditory information, you can also import audio files or videos, such as lectures, and create text-based information out of them. From our research, we have been able to determine that this may assist learners with APD (Auditory Processing Disorder)

Modern languages assessments/ tasks this tool can help with

Podcasts

Audio projects demonstrating linguistic, analytical and intercultural skills

Example prompt

"Act as an academic portfolio coach [Role]. Help me organize my coursework evidence into categories and draft reflective entries showing my linguistic growth [Task]. I am putting together a final language development portfolio [Context]."

Disclaimer: you can also create an audio overview of your sources

Portfolios

Collections of language and cultural work demonstrating development over time.

Example prompt

"Act as an academic portfolio coach [Role]. Help me organise my coursework evidence into categories and draft reflective entries showing my linguistic growth [Task]. I am putting together a final language development portfolio [Context]."

Disclaimer: uploading all your sources/ work and prompting in this fashion is most effective for this task.

Vocabulary tests

Evaluate understanding and use of vocabulary, idioms and language registers appropriate to different contexts

Example prompt

"Act as a Spanish language instructor [Role]. Create a flashcard set and a self-test quiz using these 30 vocabulary items from my recent seminar: [paste words] [Task]. I am a first-year university student revising for an upcoming module vocabulary test [Context]."

Disclaimer: uploading all your sources/ work and prompting in this fashion is most effective for this task.

Desktop

Setting up/ signing in

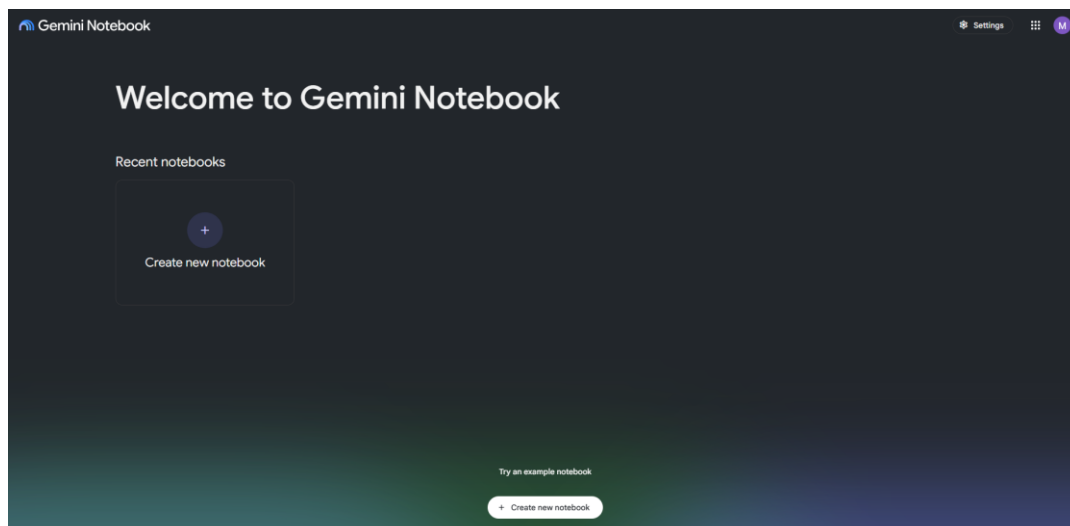
How to do this

You can access Gemini Notebook but simply searching “Gemini Notebook” into Google Chrome and select the first appropriate result. From that point you can sign in with your Cardiff University email address.

Disclaimer: Users are strongly advised not to input personal, sensitive, confidential, or identifiable information into AI tools. Only information that can be shared safely and ethically should be used, in accordance with institutional policies and relevant data protection requirements.

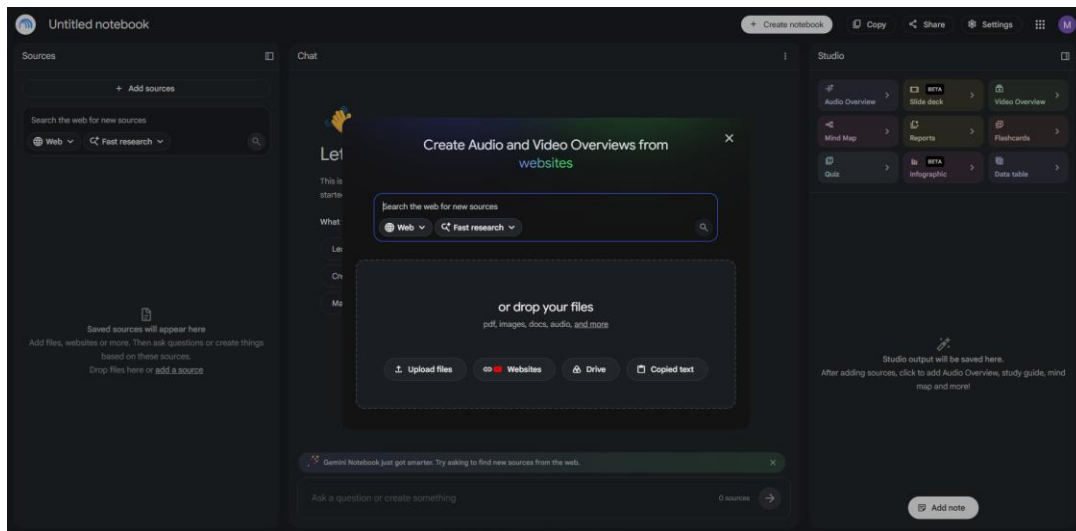
After set up/ signing in

You should be greeted with the following interface:



From this point, you have access to Gemini Notebook, provided by Cardiff University. You can create a new notebook by clicking the “Create new notebook” button.

Upon doing this, you should be presented with the following interface:



You can now upload your learning resources, via drag and drop, or by clicking the “Upload files” button in the popup. When you do this, you will be able to ask Gemini questions, and it will only give you answers based on the information it knows from what you have uploaded. You can ask questions in the “Ask a question or create something” bar.

Summarising long pdfs/ documents

Gemini Notebook allows you to upload long pdfs and documents, as the model's source of information, this means that Gemini Notebook can summarise pdfs and documents effectively.

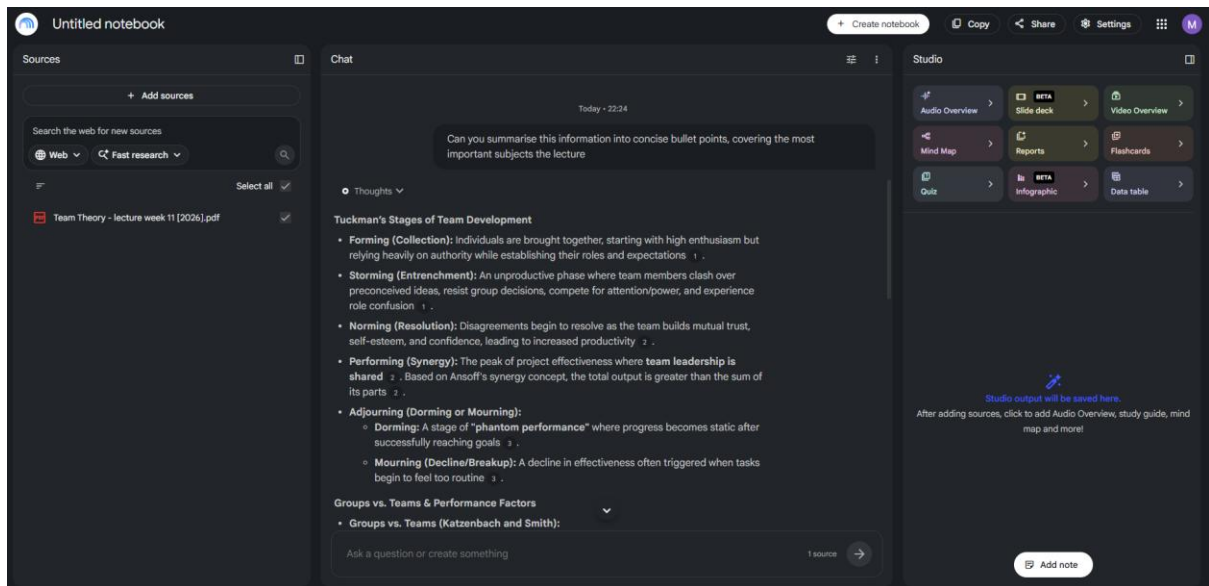
How to do this

To demonstrate this, lecture material, from the School of Computer Science and Informatics (produced by Dr Daniel J Finnegan) has been uploaded to Gemini. Gemini will be requested to summarise this source of information into concise bullet points.

An example prompt to do this is as follows:

“Can you summarise this information into concise bullet points, covering the most important subjects the lecture”

Here is what happens when this example prompt is applied to this lecture material in Gemini Notebook:



Gemini Notebook was able to effetely summarise and cover the most important parts of the lecture material.

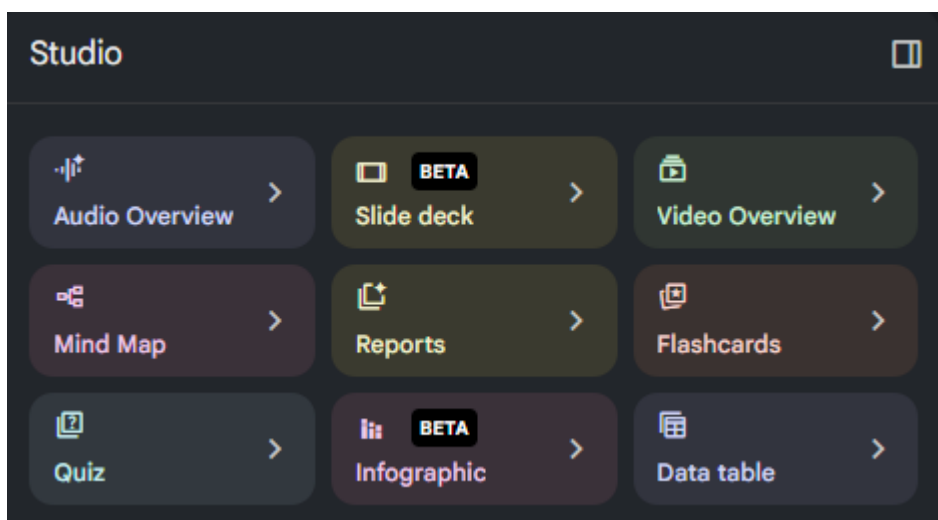
As you can see, there is now a list of sources on the left side of the interface. You can add more sources, or disable them, stopping Gemini from using them as a reference. You can add sources by clicking the “+ Add sources” button near the top of the interface on the left side.

Create quizzes and audio files from sources uploaded

Other than being able to use just what you upload to provide information, Gemini Notebook can also produce different learning recourses such as quizzes, and video overviews.

How to do this

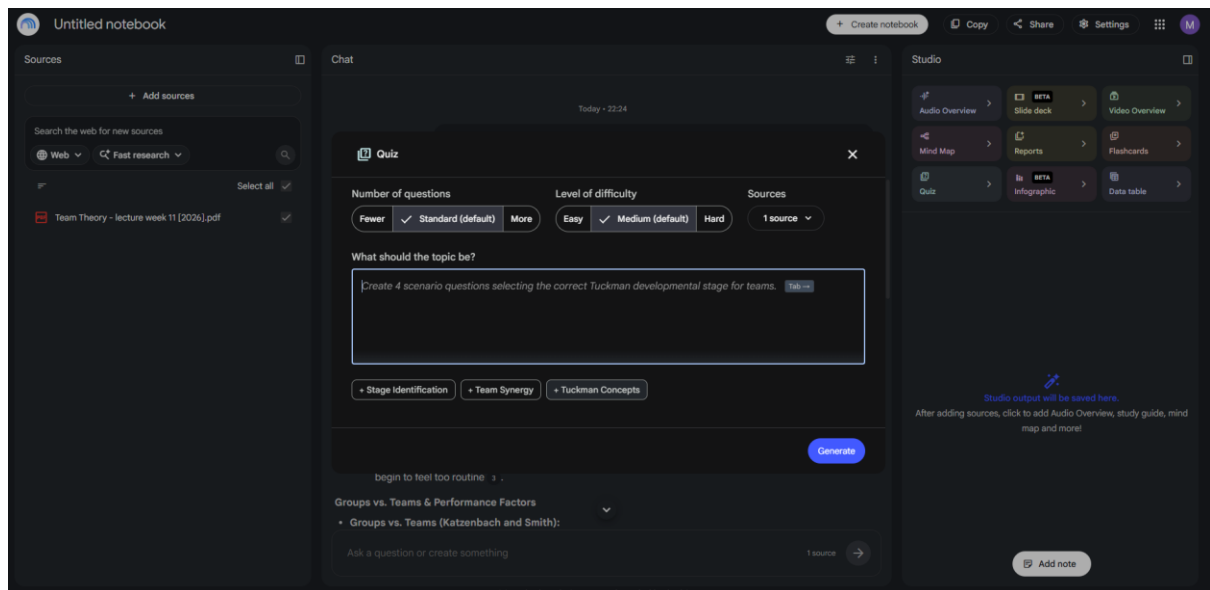
Below is the list of different recourses that Gemini can create:



Student Guide | Gemini Notebook

These can be seen on the right side of the interface. With the same lecture material uploaded, it will be used to demonstrate how Gemini Notebook produces quizzes

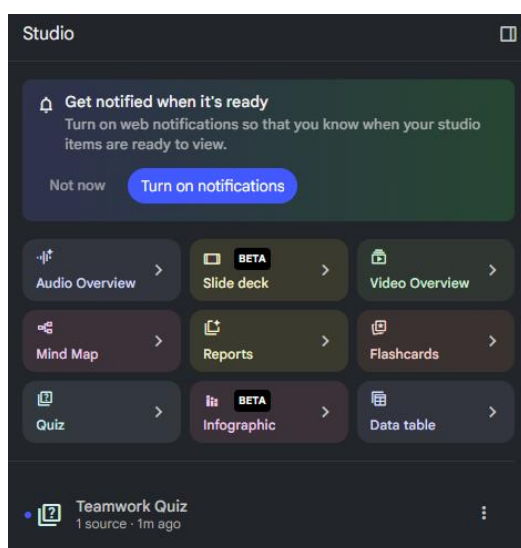
The following is the result of this:



A menu has appeared, showing the customisability options that there are for this quiz feature. Select and prepare topics based on what your desired task requires.

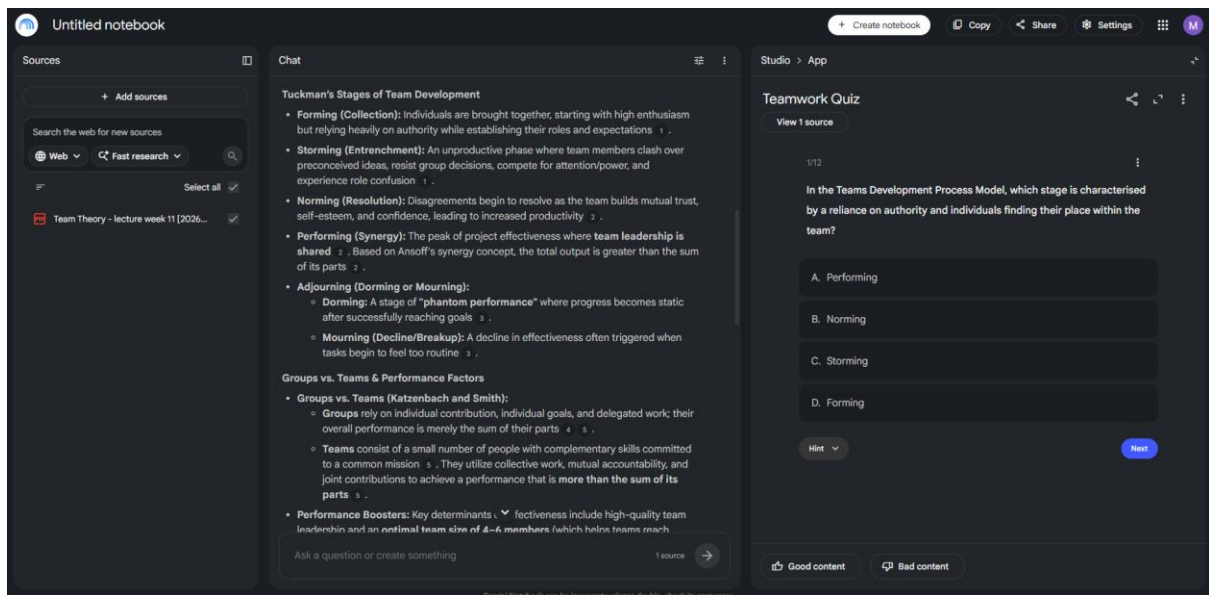
Upon clicking the “Generate” button, the quiz generation will begin and will be accessible when finished. It will load below the menu on the right side of the screen, below the table of different recourses to generate.

Once the quiz is finished generating, you will see something like this in that area of the interface:



You can then click on the quiz, and it will begin.

For reference:



On the right side of the interface, the quiz has begun, and the generation of this learning recourse is complete. You can edit, change and reshape this recourse and other recourses, even after it has been generated already.

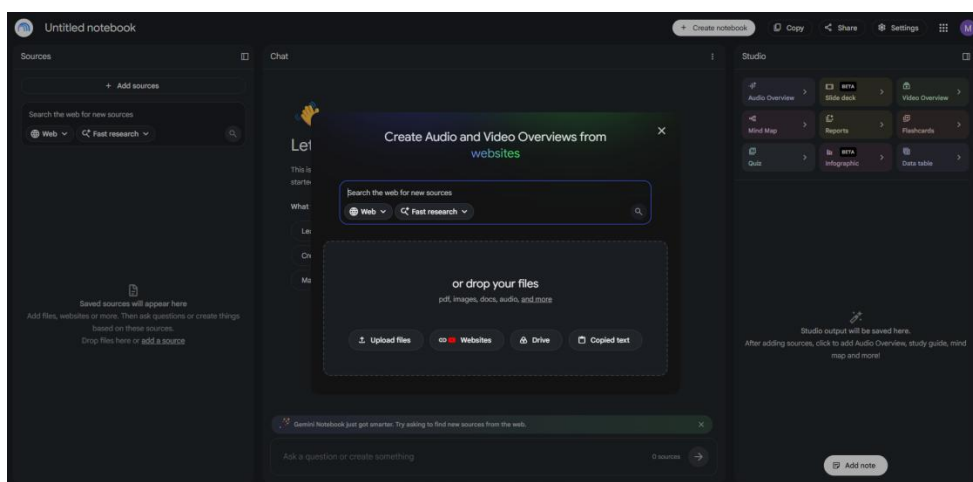
Uploading lectures, using them as sources and transcribing them

Gemini Notebook also lets you upload audio and video files, as does the normal version of Gemini, but it also lets you paste in YouTube videos, and lets you use all the above as a source.

How to do this

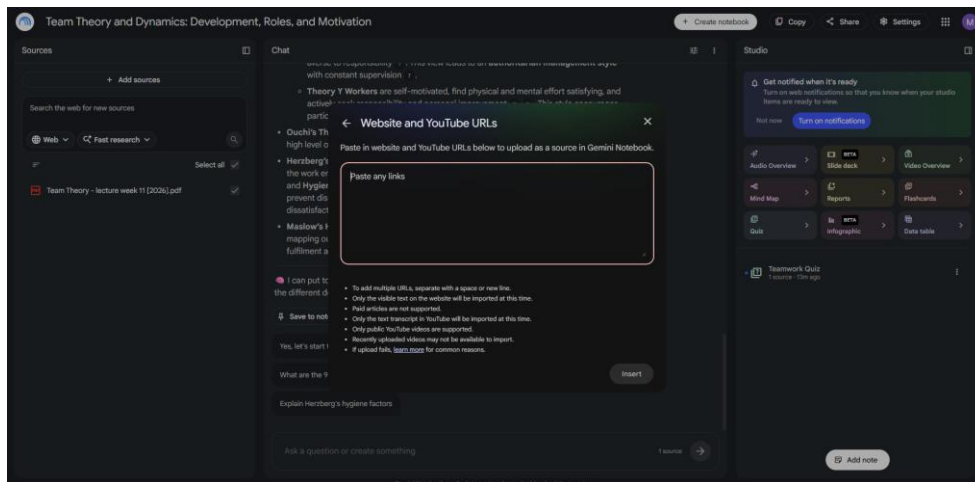
To demonstrate this, a YouTube video will be pasted into Gemini Notebook. This can also be done with .mp3 and .mp4 files. To get to the part of the interface that allows you to do this, select the “+ Add sources” button, near the top left corner of the interface.

You should be greeted with the following interface:



From this point, select the “Websites” button

You should be greeted with the following interface:

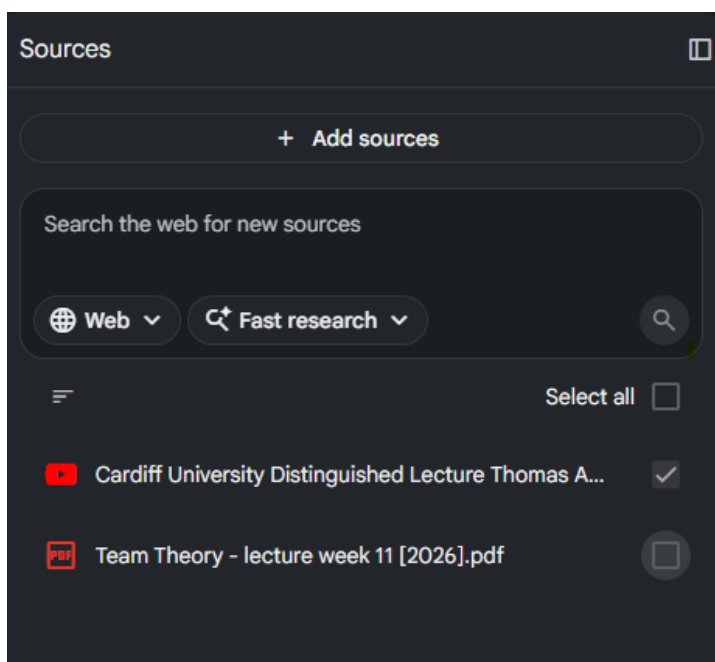


At this point you need to get the link of the YouTube video that you want to use as a source and paste it into this box, and after that, click the “Insert” button.

To demonstrate how this can be done, a lecture posted by Cardiff University on YouTube (“**Cardiff University Distinguished Lecture Thomas Andriacchi: Why No Treatment for Osteoarthritis?**”) will be uploaded to Gemini Notebook.

With these two sources uploaded, and only wanting to use the video, I will be disabling the use of the initial source.

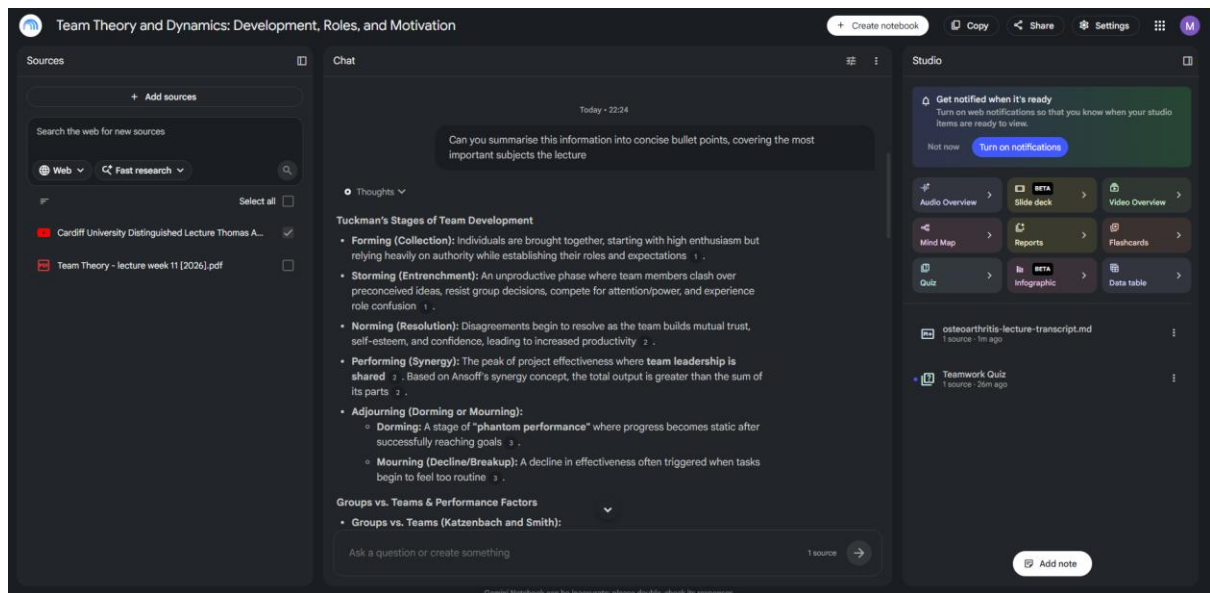
For reference:



Here is an example prompt that can demonstrate uploading lectures, using them as a source, and transcribing them:

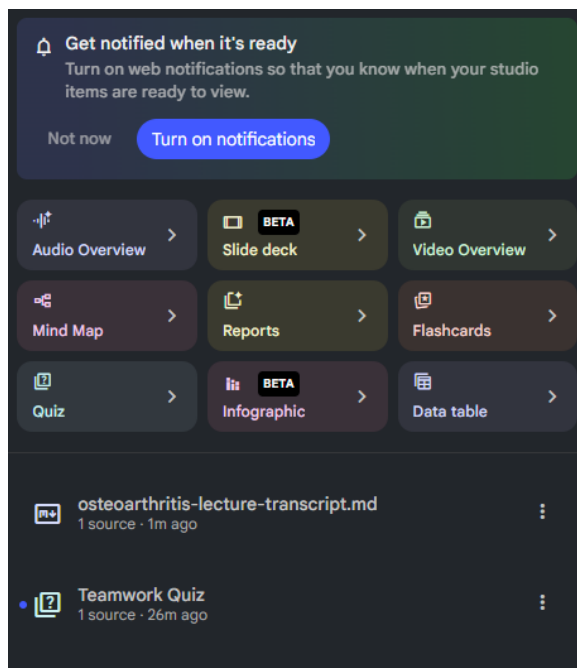
“Please can you transcribe this YouTube video, with a summary of the most important points”

Here is the result of using that example prompt in this situation:

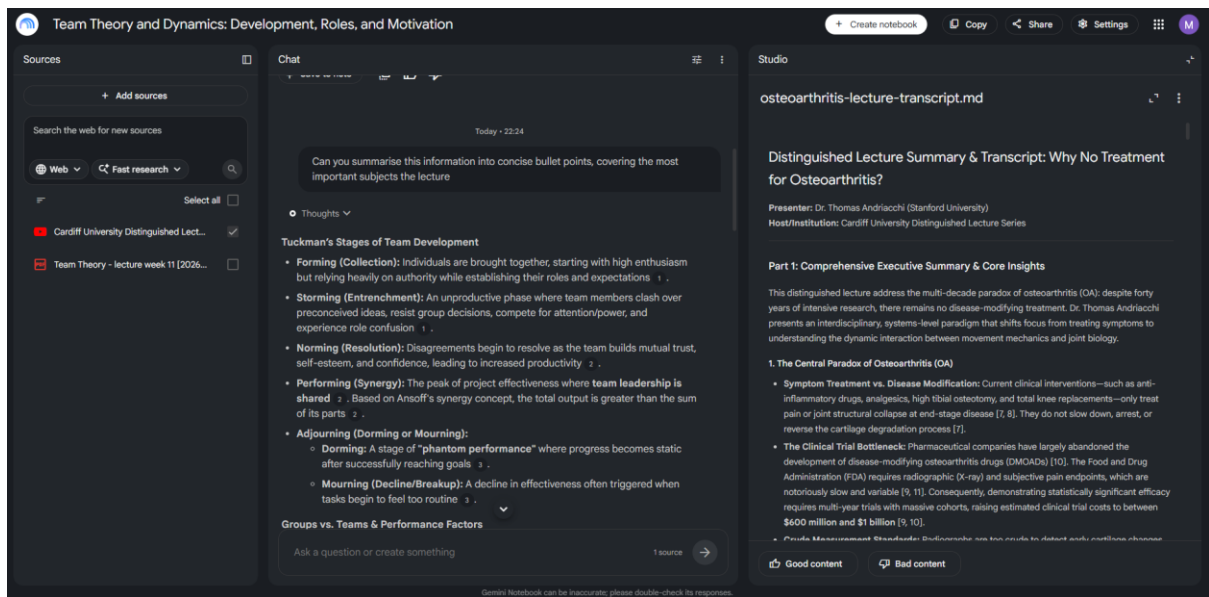


Gemini Notebook generated the most important points into the main conversation space, in the middle of the interface, but also produced a full transcript document, accessible in the recourse area, in which the previously mentioned quiz is also accessible.

For reference:



Upon clicking the transcript document:



Gemini notebook was able to give a summary of the most important parts of the lecture, and then a full transcript that is fully accessible in the “Studio” section that can be viewed and scrolled along with viewing other parts of the interface.

Further recourses (Provided by Cardiff University)

Further guidance for using Gemini Notebook

<https://intranet.cardiff.ac.uk/students/applications/view/gemini>

References

Child Mind Institute (2026) What is neurodiversity? Available at:

<https://childmind.org/article/what-is-neurodiversity/> (Accessed: 3 September 2026).

Eves, L. (2026) Neuroinclusive teaching in higher education. Higher Education Policy Institute. Available at: <https://www.hepi.ac.uk/2026/01/26/neuroinclusive-teaching-in-higher-education/> (Accessed: 3 September 2026).

For Dyslexia

University of Strathclyde (n.d.) Barriers faced by students with dyslexia. Available at: <https://www.strath.ac.uk/professionalservices/disabilityandwellbeing/informationforstaff/workingwithstudentswithdyslexia/barriersfacedbystudentswithdyslexia/> (Accessed: 2 July 2026).

For Autism

Reachout ASC (n.d.) Autism and behaviour in secondary school. Available at:

<https://reachoutasc.com/autism-and-behaviour-in-secondary-school/>

(Accessed: 2 July 2026).

For processing disorders

National Center for Learning Disabilities (n.d.) Visual and auditory processing disorders.

Available at: <https://www.ldonline.org/ld-topics/processing-deficits/visual-and-auditory-processing-disorders> (Accessed: 10 September 2026)

For Dyspraxia

Dyspraxia Foundation (2012) Dyspraxia in further and higher education. Available at:

<https://dyspraxiafoundation.co.uk/wp-content/uploads/2014/12/Dyspraxia-in-Further-and-Higher-Education-revised-2012.pdf> (Accessed: 10 September 2026).

For ADHD

Royal Borough of Windsor and Maidenhead (n.d.) Barriers to learning: poor attention and concentration and/or hyperactivity (ADHD). AfC Info. Available at:

[https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-](https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-)

[advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-](https://rbwm.afcinfo.org.uk/pages/local-offer/information-and-advice/education/about-sen-support-in-schools/ordinarily-available-provision/barriers-)

to-learning/barrier-to-learning-poor-attention-and-concentration-and-or-hyperactivity-adhd (Accessed: 10 September 2026).

Cardiff University Support services

<https://www.cardiff.ac.uk/study/student-life/student-support/disability-and-dyslexia-service>